Training Vectors / Test Vectors (QP Formulation)

Use C=1

Training Vectors

x1	x2	x3	x4	
T1	4.5	12.5	-72	-84.5
T2	12.5	40.5	-264.5	-312.5
T3	-72	-264.5	2178	2664.5
X1	-84.5	-312.5	2664.5	3280.5

X1 X2 x3 x4

Test Vectors

Minimize:

$L_D(\alpha) = \frac{1}{2} \sum_{j=1}^{N} \sum_{i=1}^{N} \alpha_i \alpha_j y_i y_j [X_i^T X_j + 1]^2 - \sum_{i=1}^{N} \alpha_i$

$= \frac{1}{2} [\alpha1\ \alpha2\ \alpha3\ \alpha4] \cdot Q \cdot [\alpha1; \alpha2; \alpha3; \alpha4] - [\alpha1+\alpha2+\alpha3+\alpha4]$

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4.5    12.5   -72    -84.5
12.5   40.5   -264.5 -312.5
-72    -264.5 2178   2664.5
-84.5  -312.5 2664.5 3280.5
```

Subject to:

$\sum \alpha_i y_i = 0$

$\alpha1 + \alpha2 - \alpha3 - \alpha4 = 0$

$\alpha1 \le 1, \alpha2 \le 1, \alpha3 \le 1, \alpha4 \le 1$

$\alpha_i \ge 0 \quad (i = 1..4)$